

Biology
(2025)

Biotech Techniques

works because DNA is universal
(all DNA have same bases, sugar and phosphate group)

12.1.1 Definition of Biotechnology

> **Traditional**: harnessing of the potential of processes (eg: fermentation)

> **Modern**: genes manipulated to form a product or a feature which is beneficial for humans.
genetic engineering, tissue culture, mutagenesis.

12.2.1 Principles (restriction endonuclease, EcoRI, ligase, recombinant DNA)

5 steps to form one such as insulin

12.2.2 Gene tech techniques

DNA profiling: 5 steps

- ① cleave/isolate gene of interest
> restriction enzymes
- ② insert into vector
> plasmid
> recombinant DNA
- ③ insert into host
> EcoRI
- ④ check which host cell uptake recombinant vector

① Isolating gene

✓ restriction enzymes
cleave the sugar-phosphate backbone
+ direct synthesis in lab
· very new tech
+ reverse transcription
make DNA from viral RNA with
retroviruses and reverse
transcriptase
Recombinant DNA is cloned

② inserts to vector

> plasmid: extrachromo. circular DNA involved in sex. repr. + antibiotic resistance
(symbiotic relationship with host cell)
E. coli.

> restriction enzymes *EcoRI* found in *Escherichia Coli*. cleave at particular
restriction sites: between G and A in palindromic sequence GAATTC.
produces sticky ends (single-stranded DNA). → anneal H-bonds
cleaves both gene of interest and vector for complementary base pairing

> ligase anneals to form recombinant DNA by forming missing phosphodiester bonds

> bacteriophage λ (lambda) vectors: viruses that infect bacteria.

③ introduce to host cell

incorporate into living cells to be replicated and expressed.

Other methods:

- > transformation: uptake of genetic material from env. by bacterial cell
- > transfection: the process by which foreign DNA is introduced to eukaryotic cells
- > microprojection
- > electroporation (apply voltage to cell to increase permeability of membrane)
- > micro injection

④ select transformed cells ✓ genetic screening ✓ replica plating

> 3 possibilities

↳ ✓ nitrocellulose paper

② no plasmid uptake (untransformed cells) (99%)

③ recombinant plasmid uptake

④ non-recombinant plasmid uptake (its own DNA was annealed instead of the foreign GoI)

> marker genes, DNA probe

> 2 marker genes: tetracycline and ampicillin.

② will die with tetracycline, while ③ ④ have resistance in plasmid.

④ will die with ampicillin, as foreign gene was inserted into the middle of the marker gene.

> replica plating: before adding ampicillin, a copy of the petri plate (marker plate) is taken by nitrocellulose paper to make. The ④ that die can be located on the copy

⑤ Grow + collect:

insulin will be produced in high quantity and pure form.

> PCR Polymerase Chain Reaction

① length of DNA mixed with free nucleotides, DNA polymerase.

② heat to 95°C (break H-bonds, strands separate)

③ cool to 40-60°C + add primers to anneal with complementary DNA sequence.

primers are ~20 nucleotides long & complementary to ends of target DNA.

④ raise to 72°C for tag polymerase optimal function

it attaches to primer and adds nucleotides.

⑤ after 4 cycles, half consists of target DNA and half of flanking.

12.2.3 Gene tech analytical tech ("other techniques")

> electrophoresis: separation of DNA fragments according to size
after being cleaved with restriction enzyme into segments.
high voltage electric current → attracts (-) charge DNA to (+) electrode.
↓ size / log₁₀ molecular weight, ↑ movement.
probe added to visualize DNA with UV light by intercalation (insertion)

> Southern blotting & gene probes: detection of fragments
after cleave with restriction enzyme + electrophoresis
transferred to blotting paper membrane
incubated with probe (single-stranded DNA)

> radioactive or fluorescent or enzymically
probe anneals to complementary sequence, so fragment stands out
exposed to X-ray film, UV or chemiluminescence.

> PCR:

Practical work

take pancreas → homogenize → digest/enzymes → DNA spooled
lyses cell membranes or centrifuged
+ nuclear membranes for collection
to free DNA
EcoRI
→ restriction endonuclease
↓
← gel electrophoresis

12.2.4 Applications of Biotech

Genetic fingerprinting + paternity tests
VNTR (variable number tandem repeat)

Pharmaceutical projects:

> human protein replacement

> insulin

> human growth hormone

> modern vs traditional

Agricultural animals

> antithrombin (anticoagulant + anti-inflammatory properties)
for hereditary antithrombin deficiency

Gene therapy

> X-linked SCID (severe combined immunodeficiency)
bone marrow stem cells

> ethics

Agriculture: Bt maize, pest-resistant crop

> env. implications

		using radioactive in situ hybridization is not examinable).
		Practical work should include precipitation and spooling of DNA and gel electrophoresis of DNA fragments.
12.2.4 Applications of gene technology		Knowledge of the following applications is needed though details of the specific processes are not required. Genetic fingerprinting and DNA profiling and its application in forensic work and paternity cases. Pharmaceutical products of gene technology: human protein replacement as exemplified by the production of either insulin or human growth hormone (somatropin or somatotrophin) by genetically modified microorganisms; advantages over traditional methods of treatment (e.g. somatropin from farm animals might result in persons developing CJD up to 30 years after receiving hormonal treatment) Use of agricultural animals in transgenic technology as exemplified by genetically modified goats to produce a recombinant form of human antithrombin (ATryn®), having anticoagulant and anti-inflammatory

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		phage and cosmid vectors are not needed); Introducing vector DNA in the host cell; selecting the transformed cells by using marker genes and an adequate DNA probe – (in this context the terms genetic selection, genetic screening and replica plating should be used). Other methods of introducing foreign DNA in host cells, e.g. transformation, transfection, microprojectiles, electroporation and microinjection, are not expected except in outline only where relevant for understanding of applications listed below. The polymerase chain reaction (PCR) as a method of producing multiple copies of a particular gene.
12.2.3 Analytical techniques of gene technology		Methods of analysing DNA organisation: separation of DNA fragments by gel electrophoresis; detection of fragments using Southern blotting and radioactive gene probes; (localisation of genes on chromosomes using radioactive in situ hybridization is not examinable). Practical work should include precipitation and spooling of DNA and gel electrophoresis of DNA fragments.
12.2.4 Applications of gene technology		Knowledge of the following applications is needed though details of the specific processes are not required. Genetic fingerprinting and DNA profiling and its application in forensic work and paternity cases. Pharmaceutical products of gene technology: human protein replacement as exemplified by the production of either insulin or human growth hormone (somatropin or somatotrophin) by genetically modified microorganisms; advantages over traditional methods of treatment (e.g. somatropin from farm animals might result in persons developing CJD up to 30 years after receiving hormonal treatment) Use of agricultural animals in transgenic technology as exemplified by genetically modified goats to produce a recombinant form of human antithrombin (ATryn®), having anticoagulant and anti-inflammatory

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AM Syllabus (2025): Biology

		properties for the treatment of patients with hereditary antithrombin deficiency. Gene therapy: Use of bone marrow stem cells in the treatment of X-linked SCID (severe combined immuno-deficiency). Awareness of the limited success of treatment obtained so far with gene therapy. Applications of gene technology in agriculture: the production of pest resistant crops e.g. Bt maize. Note: Alternatives to the traditional textbook examples of gene technology are being suggested as gene therapy for cystic fibrosis has had very limited success: the use of bovine somatotropin (BST) in dairy herds is not approved in the EU. Candidates should be able to evaluate briefly the environmental implications of GM plants and ethical implications of the use of stem cells and gene therapy.
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what is bovine somatotrophin (BST)?

Essay: describe techniques involved in Biotech [25]



